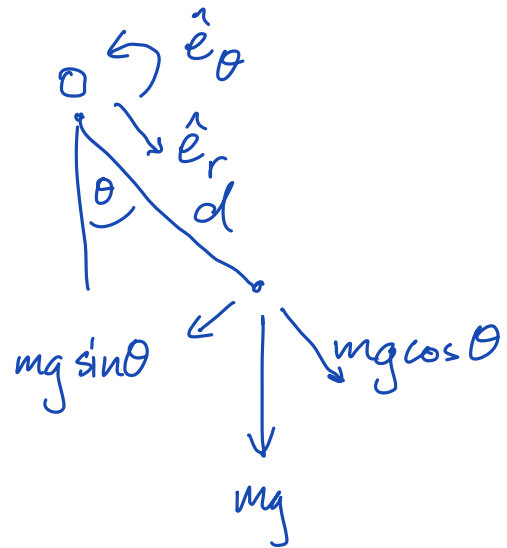
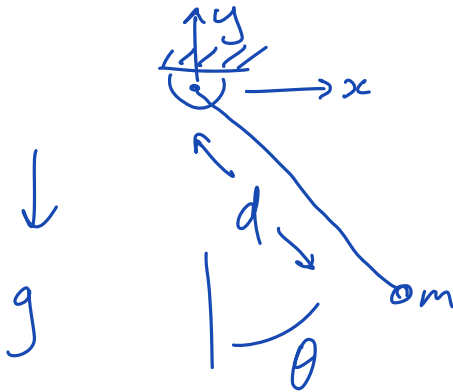


Simple Pendulum

- Minimal Coords
- Maximal Coords
- DAE in Matlab

Prelim | Wed/Thurs.

Setup.



Minimal Coordinates Approach

Finding Equation for $\ddot{\theta}$ using N-E

AMB @ O. $\sum M_{/O} = \dot{H}_{/O}$

$$I_O = \cancel{I_C} + md^2$$

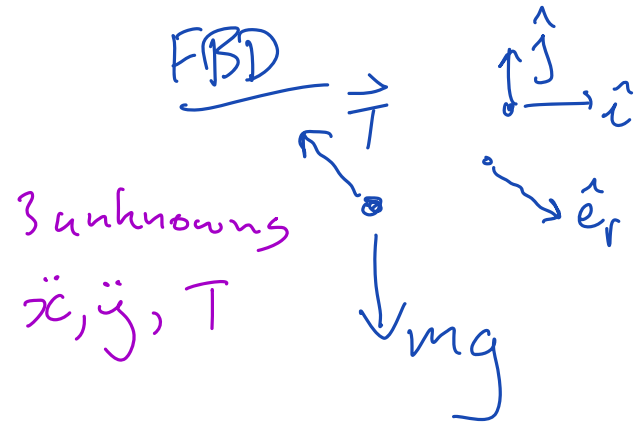
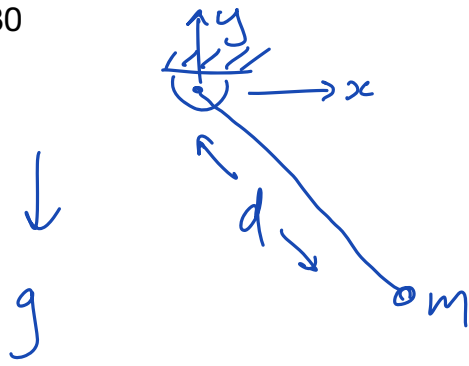
$$-mgd \sin \theta = \frac{d}{dt} (\vec{\omega} I_O)$$

$$= \frac{d}{dt} (\dot{\theta} md^2)$$

$$\cancel{-mgd \sin \theta} = \cancel{\ddot{\theta} md^2}$$

$$\ddot{\theta} = -\frac{g}{d} \sin \theta$$

(Try xy)



3 unknowns
 $\ddot{x}, \ddot{y}, T$

Maximal Coordinate Approach

DAE "Differential Algebraic Equation" Approach - Ruina

LMB $\sum \vec{F}_{\text{ext}} = m\vec{a}$

$$-F_g \hat{j} - T \hat{e}_r = m(\ddot{x} \hat{i} + \ddot{y} \hat{j})$$

$$\hat{e}_r = \frac{x \hat{i} + y \hat{j}}{\sqrt{x^2 + y^2}}$$

$$\hat{i} \Rightarrow \frac{-T x \hat{i}}{\sqrt{x^2 + y^2}} = m \ddot{x} \hat{i}$$

$$m \ddot{x} + T \frac{x}{d} = 0 \quad (1)$$

$$\hat{j} = -mg \hat{j} - \frac{T y \hat{j}}{\sqrt{x^2 + y^2}} = m \ddot{y} \hat{j}$$

$$m \ddot{y} + T \frac{y}{d} = -mg \quad (2)$$

Constraint Equation

$$\sqrt{x^2 + y^2} = d$$

$$x^2 + y^2 = d^2$$

$$\frac{d}{dt} \Rightarrow 2\dot{x}\dot{x} + 2\dot{y}\dot{y} = 0$$

$$\frac{d}{dt} \Rightarrow 2\ddot{x}x + 2\dot{x}^2 + 2\ddot{y}y + 2\dot{y}^2 = 0$$

$$x\ddot{x} + y\ddot{y} = -(\dot{x}^2 + \dot{y}^2) \quad (3)$$

Hints

- Get rid of " $\sqrt{\quad}$ " if possible.
- find expression in terms of $\ddot{x}, \ddot{y}$

Solve 3 equations

$$\underbrace{\begin{bmatrix} m & 0 \\ 0 & m \\ x & y \end{bmatrix}}_A \underbrace{\begin{bmatrix} \ddot{x} \\ \ddot{y} \\ T \end{bmatrix}}_u = \underbrace{\begin{bmatrix} 0 \\ -mg \\ -(\dot{x}^2 + \dot{y}^2) \end{bmatrix}}_F \quad \begin{matrix} (1) \\ (2) \\ (3) \end{matrix}$$

$$u \Rightarrow u = A^{-1}F$$

$$\text{in Matlab } u = A \setminus F$$

DAE in Matlab

function Xdot = pendulum2D(t, X, p)

m = p.m ... (etc)

r = X(1:2); v = X(3:4);

$$A = \begin{bmatrix} m & 0 & x/d \\ 0 & m & y/d \\ x & y & 0 \end{bmatrix}; \quad F = \begin{bmatrix} 0 \\ -mg \\ -v^2 \end{bmatrix}$$

$$u = A/F \rightarrow u = \begin{bmatrix} \ddot{x} \\ \ddot{y} \\ T \end{bmatrix}$$

$$vdot = u(1:2)$$

$$rdot = v$$

$$Xdot = [rdot; vdot]$$